

ARKO

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TECHNICAL SKILLS

AI Orchestration: Multi-Agent Systems, LLMOps, Model Routing, Cost Optimization, LangGraph, LangChain, MCP (Model Context Protocol), Function Calling, Prompt Engineering, Structured Output

AI Safety: Guardrails, Hallucination Detection, Input/Output Validation, PII Detection, Human-in-the-Loop, AI Evaluation/Evals

Languages: Python, TypeScript, Rust, SQL

Stack: FastAPI, Next.js, React, PostgreSQL, pgvector, Redis, Docker, AWS, Kubernetes, WebSocket/SSE, CI/CD

PROFESSIONAL EXPERIENCE

Independent AI Systems Architect 2024 – Present

Multi-Agent Orchestration & LLMOps — Production AI infrastructure with cost optimization and evaluation pipelines

Multi-Agent Orchestration Platform — Production System

- Built production multi-agent orchestration platform managing 15+ concurrent AI agents across 3 model providers (Anthropic, OpenAI, open-source) with intelligent model routing matching task complexity to cost tier
- Designed cost optimization layer reducing LLM API spend by 40% through automatic model selection — routing simple tasks to lightweight models and complex reasoning to frontier models
- Implemented real-time agent monitoring dashboard with kill switches, cost tracking per agent, token usage analytics, and performance observability across all active workflows
- Built LangGraph-based coordination patterns for agent-to-agent communication, shared memory, and hierarchical task decomposition

Enterprise RAG & AI Integration Pipeline — 50K+ Document Corpus

- Designed multi-model RAG pipeline with custom AI evaluation suite achieving 94% retrieval accuracy across 50K+ document corpus with <2s end-to-end response time
- Built automated prompt regression testing catching quality drops before production deployment, with continuous evals running against golden datasets
- Implemented guardrails layer with hallucination detection, source attribution validation, and confidence scoring on every response

AI-Powered Business Automation — Unified Agentic Workflow Engine

- Orchestrated 5 AI workflows (email triage, meeting intelligence, scheduling, content generation, follow-up detection) through unified workflow engine with human-in-the-loop checkpoints
- Built agent coordination layer where specialized agents hand off tasks with human approval gates, consolidating 5 separate tools into one orchestrated system

AI Data Specialist — Scale AI 2023 (6 months)

- Evaluated large language models through expert human feedback on reasoning, code generation, and instruction following, directly improving model quality for production deployment
- Performed AI safety red-teaming identifying critical failure modes including hallucination patterns, prompt injection vulnerabilities, and bias in production LLM outputs

Full-Stack Software Engineer — Contract 2020 – 2023

- Shipped 8 production applications with AI integrations including intelligent search, recommendation systems, and automated content generation serving 500K+ monthly active users
- Designed streaming architectures (WebSocket, SSE) for real-time AI-powered features with <100ms latency, powering live dashboards for fintech and healthcare clients